



On-line Partial Discharge Testing for Motors & Generators

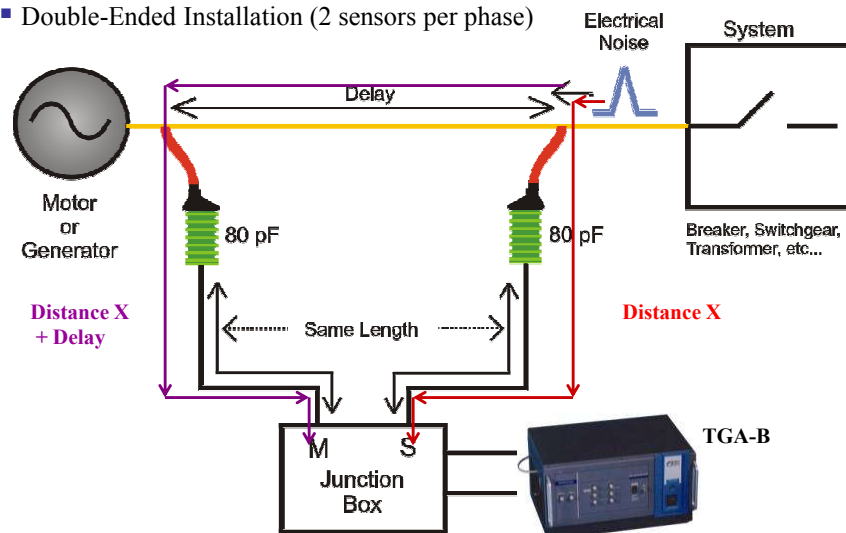
PT Dielektrika Persada Teknik
Tangerang



Directional Time-of-Arrival (Noise Separation)

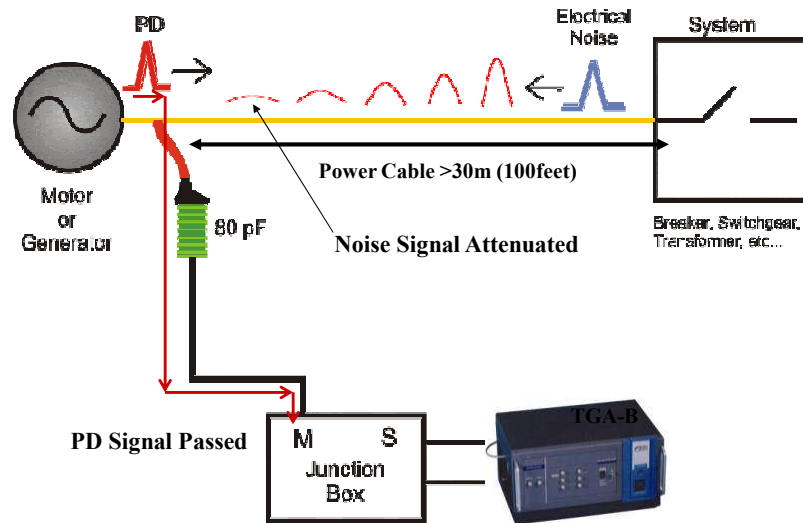


- Double-Ended Installation (2 sensors per phase)



Pulse Shape Noise Discrimination

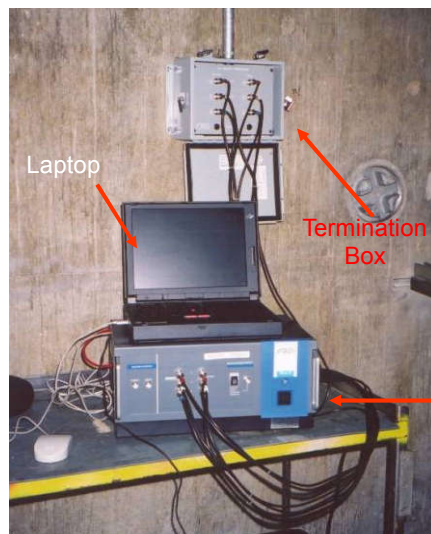
- Single-Ended Installation (1 sensor per phase)



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Periodic On-line PD Testing



Portable Instrument: TGA-B

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On-line Test Procedure



Partial Discharge Data Collection Using TGA-B

Purpose: To observe the PD that occurs in a machine during normal operations. This requires an Iris instrument and a laptop computer running PDLitePro.

Safety Precautions

1. Power supply for the instrument must be connected through GFCI device.
2. The shield of the coaxial cable should be hard grounded at the termination box. The 50/60Hz AC voltage between the center pin and the shield of PDA and Bus installs should be 100-600mVAC. If the ground is disconnected it is possible that this voltage may float to a higher value, so always check before connecting cables.



On-line Test Procedure



COLLECTION SET UP

1. Review previous partial discharge measurement and obtain database.
2. Verify database accuracy, instrument connection and instrument setup.
3. Run tester to make sure that the instrument is working correctly.



On-line Test Procedure



PARTIAL DISCHARGE DATA COLLECTION

1. Record machine operation parameters and input the data on Asset Information of PDLite software.
2. Measure voltage (mV) across BNC connection. Record values and anomalies.
3. Connect input channel on the front of the IRIS instrument to which the sensors are connected.



On-line Test Procedure



4. Setup the Iris instrument mode for the test: use BUS mode test. Be sure when using Bus mode that you set the instrument to use a C1/Mach, or Machine reference.
5. Select the specific sensitivity to run the test, or you can choose to run a sequence test. A sequence will start at the highest range (500-8500 mV) and then when directed progressively decrease until stopped. **The 2mV to 34mV and 5mV to 85mV ranges use an amplifier; so only use these when the PD levels are within these ranges to avoid damage to the Iris instrument.**
6. Set the appropriate reference angle when a measurement is performed (90°) for EMC.
7. Set appropriate delay time for correct measurement.



On-line Test Procedure



8. Run measurement and observe the quality flags.
 - Retest at higher sensitivity range if there is OVR flags.
 - Retest at lower sensitivity range if there is UND flags.
9. Verify if there are any trend changes in the PD data.
10. Verify if PD in usual position (phase shift).
11. Discussion and notes.



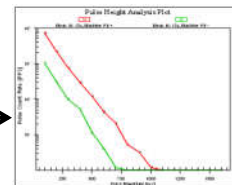
Basic Interpretation of PD Testing Results



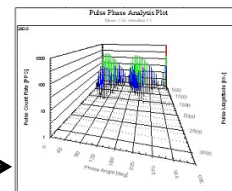
- Instrumentation provides:

1. Pulse Height Analysis Plot (2D) →

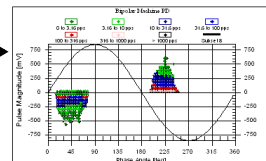
- Qm for peak PD magnitude
- NQN for total PD activity



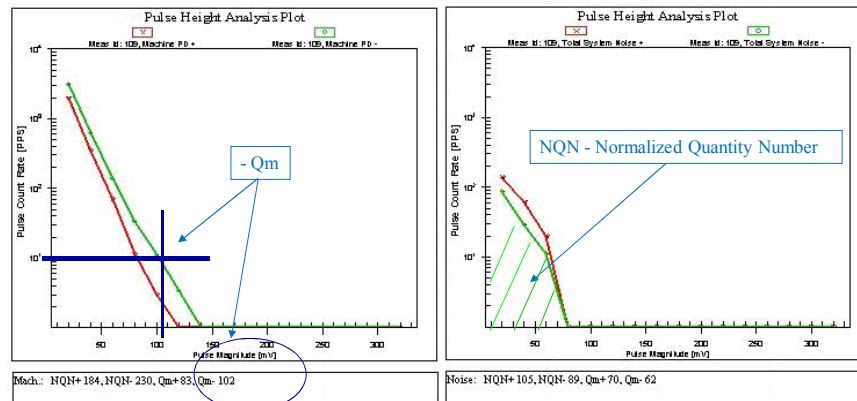
2. Pulse Phase Analysis Plot (3D) →



3. Leaner Density Plot (3D) →



Pulse Height Analysis Plot

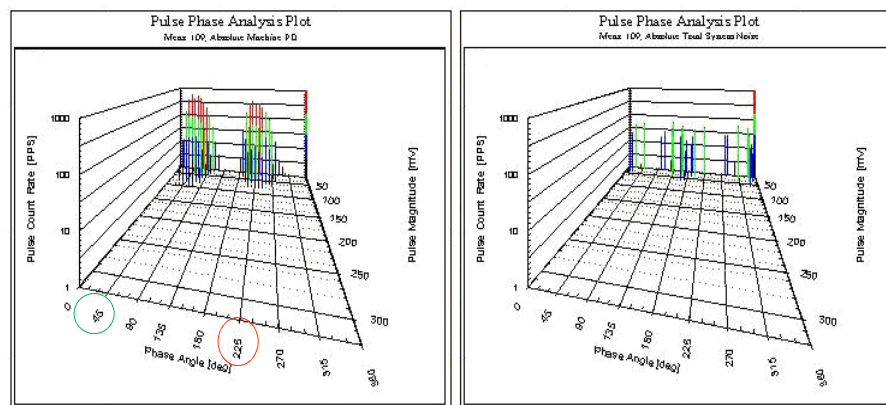


- Qm – the peak magnitude in mV at 10-pulse per second
 - Measure of void size: Qm $\uparrow \rightarrow$ Void $\uparrow$ bigger
- NQN – the representation of area beneath the curve
 - Measure of total PD activity: NQN $\uparrow \rightarrow$ Voids $\uparrow$ more

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Phase Angle Analysis Plot

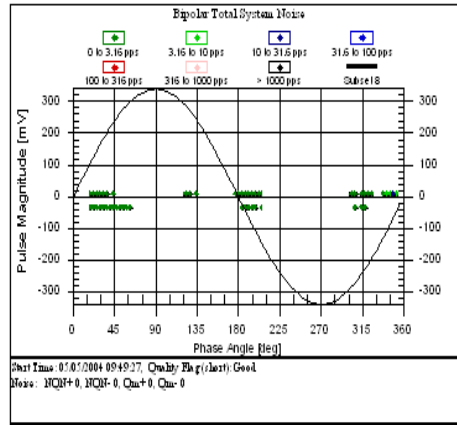
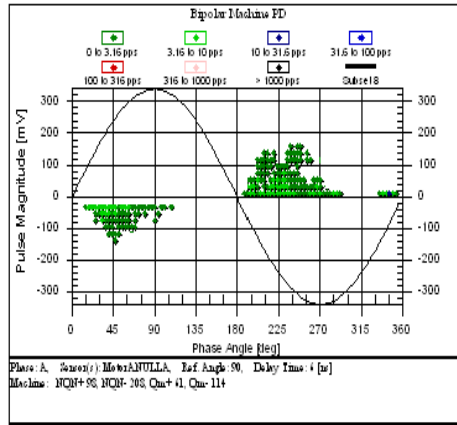


1. PD typically located at 45° and 225° with respect to the AC Phase-to-Ground voltage
2. Classic PD: – PD @ 45° & + PD @ 225°

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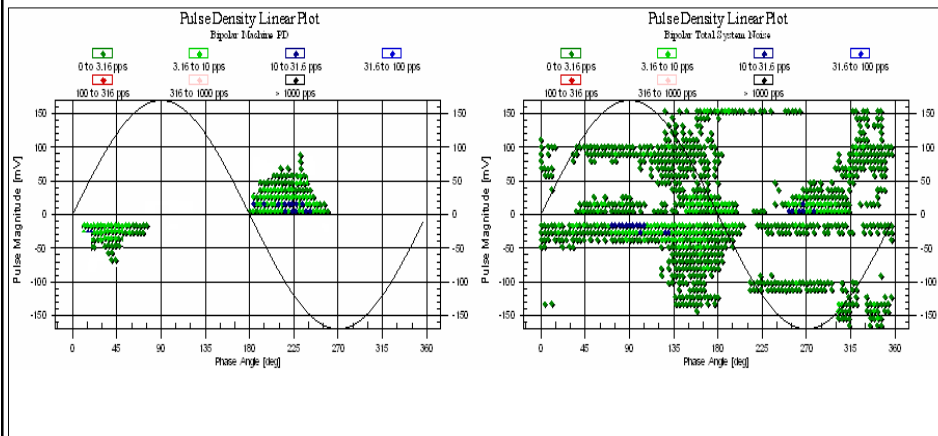
Linear Density Plot



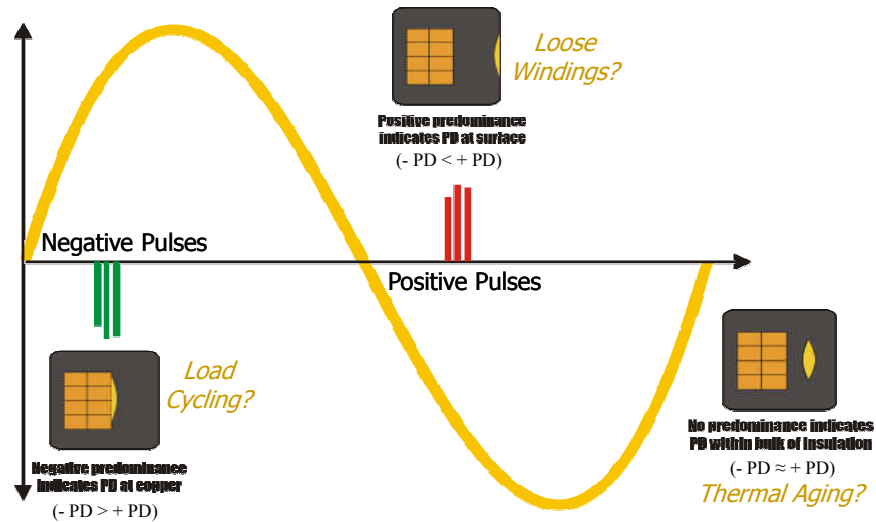
Phase Position versus Pulse Count & Magnitude



Importance of Noise Separation



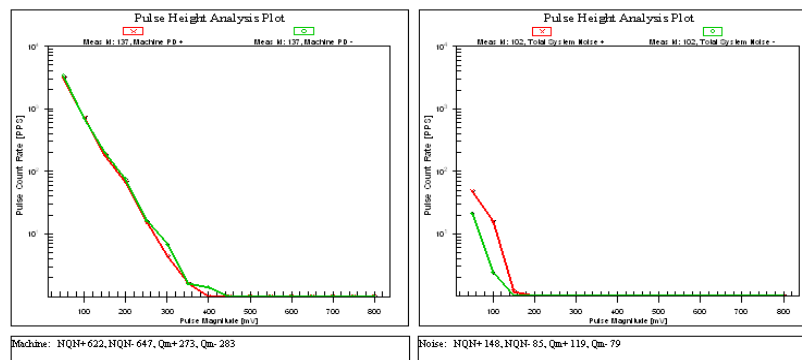
PD Pulse Polarity



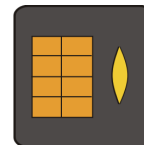
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No polarity predominance



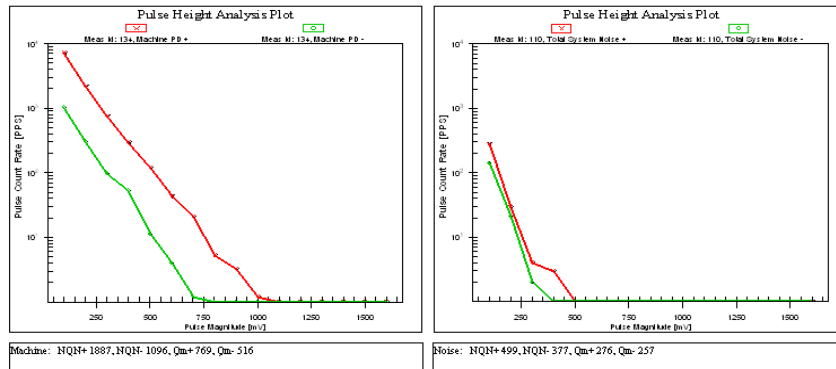
- General interpretation rules
 - No polarity predominance (+PD ~ -PD)
 - Internal voids
 - Thermal deterioration
 - Poor impregnation



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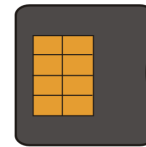
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Positive polarity predominance



General interpretation rules

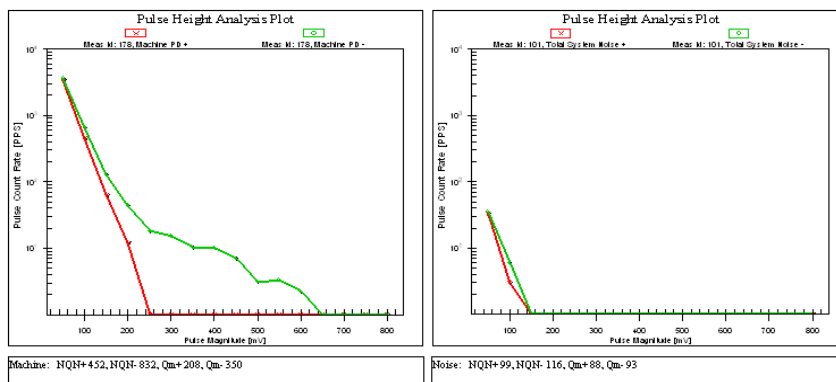
- Positive polarity predominance (+PD > -PD)
 - Discharge on surface of coils
 - Loose windings
 - Deterioration of voltage stress coatings



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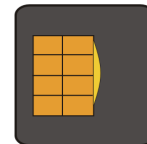
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Negative polarity predominance



General interpretation rules

- Negative polarity predominance (+PD < -PD)
 - Discharge near the copper conductor
 - Load cycling
 - Overheated develops voids near the copper conductors



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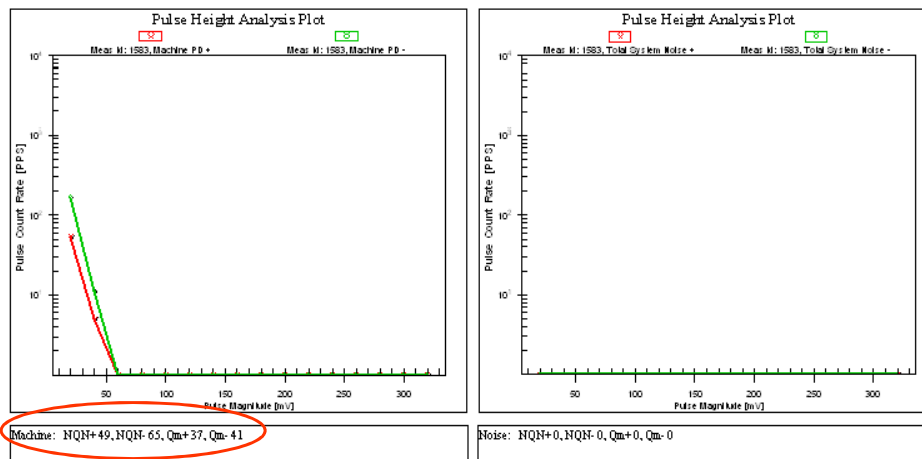
Three Basic Methods for Interpretation



- Compare graphs from identical machine (stator with highest PD is most deteriorated)
- Compare test result from a machine to a similar machines in Iris database
- Trend PD over time (2 times increase per year is significant rate of deterioration)



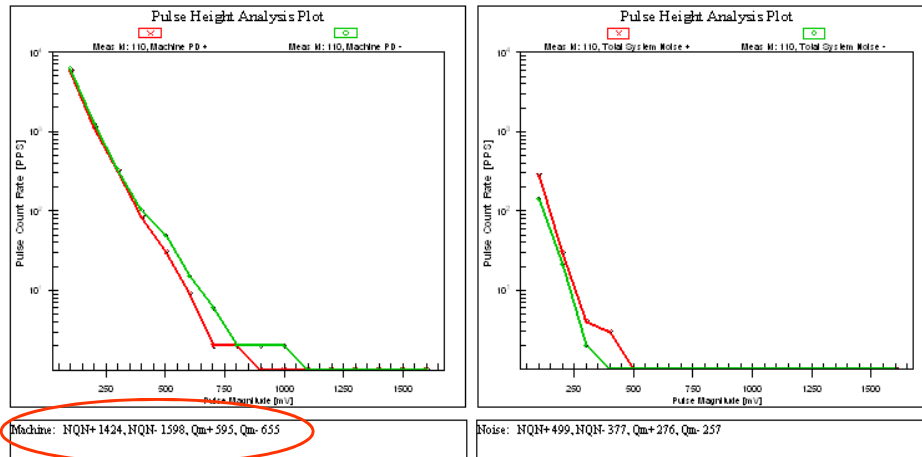
Compare Graphs from Identical Machine



A Stator with Low PD



Compare Graphs from Identical Machine



A Stator in Same Plant with Very High PD



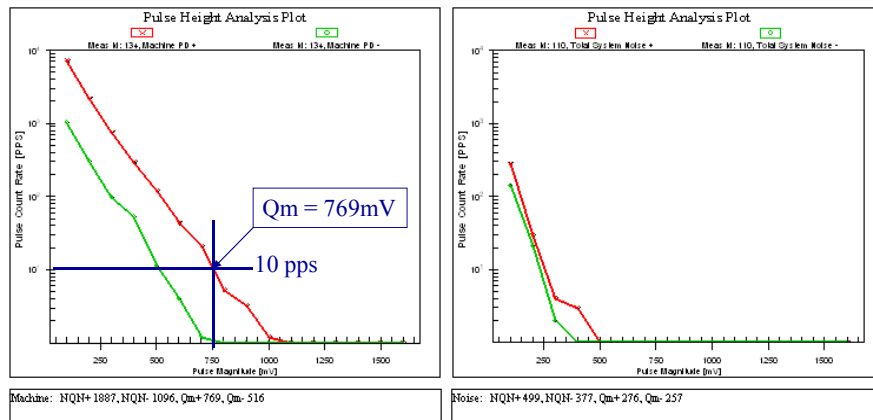
Compare Test Result with Similar Machines



- Iris PD Database
 - Over 272,000 test results from thousands of machines for comparison purposes
 - Each year Iris publishes the statistical range of Qm (peak PD) for each type of stator
 - Voltage rating
 - Air or hydrogen cooling
 - PD sensor type
 - If a stator has a Qm that exceeds 90% of readings, then winding is deteriorated



Compare Test Result with Similar Machines



Sample PD Result / 15kV Generator

- Maximum PD magnitude as read off from the 10 pps
- The Qm exceeds 90% of readings, winding is seriously deteriorated



Compare Test Result with Similar Machines



Comparison with Similar Machines

Level	% Tile	6-9 kV	10-12 kV	13-15 kV	16-18 kV	> 19 kV
Negligible	< 25th	5	24	24	34	120
Low	< 50th	38	63	99	155	179
Typical	< 75th	101	138	216	281	318
Moderate	< 90th	243	270	407	453	658
High	< 95th	652	442	688	676	1492
Very High	> 95th	652	442	688	676	1492
	Avg	116	133	195	204	337
	Max	1600	3410	3548	1630	3552

Qm Values for Bus and PDA Couplers Measured with a PDA-IV and TGA [2000]

(Air-cooled machines)



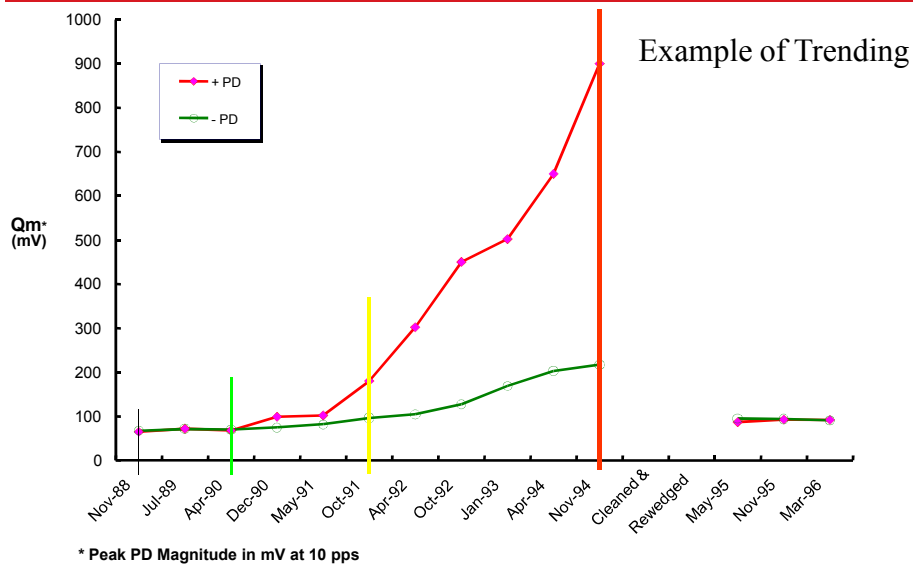
Trend PD Over Time



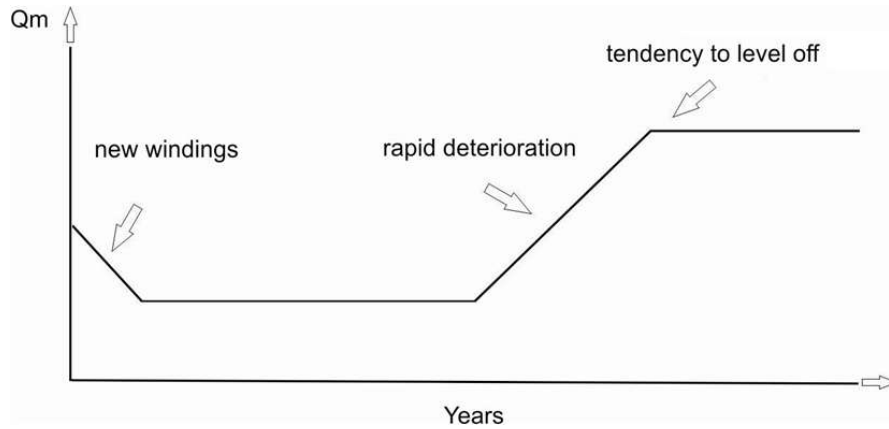
- Trending of data
 - Most powerful, reliable means of interpreting data
 - If Qm low, and constant trend, winding in good condition
 - Collect data over the years, if Qm increases slowly, winding is stable
 - If Qm doubles in 6 to 12 months, rate of deterioration increasing
 - Trending Can also determine if repairs effective



Trend PD Over Time



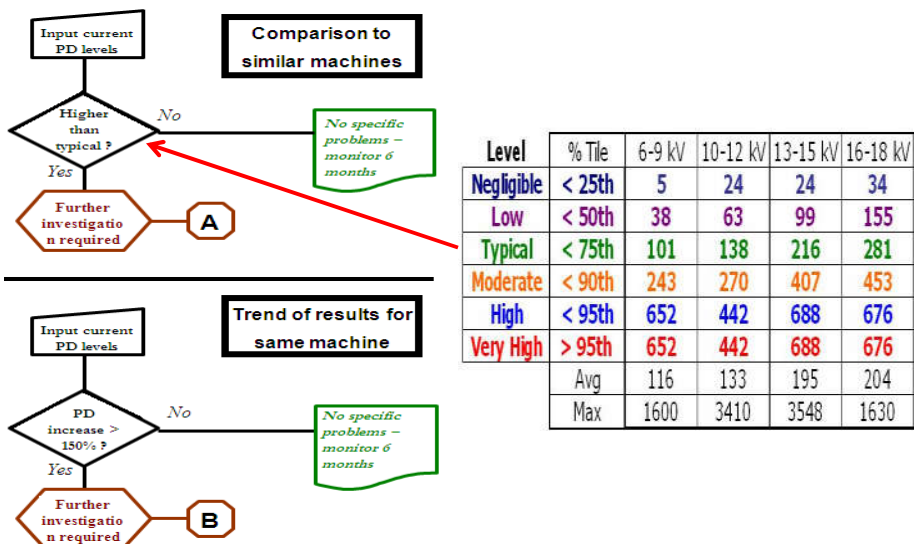
Typical Trend in PD over Winding Life



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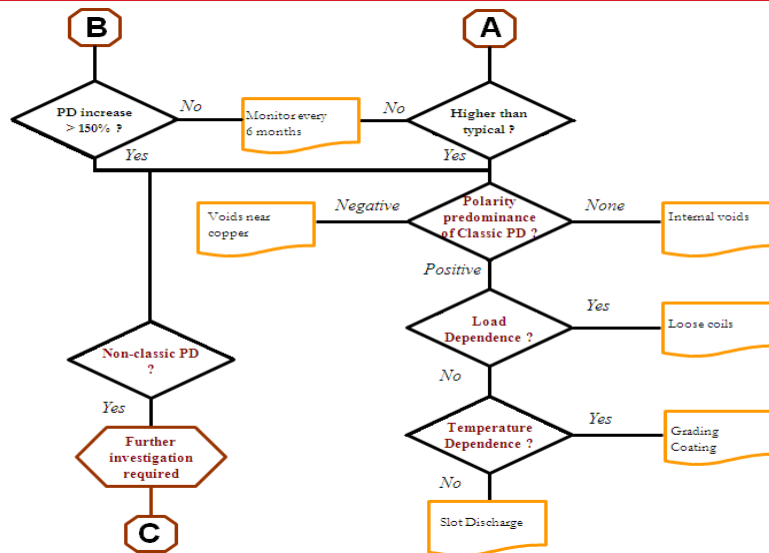
PD Interpretation



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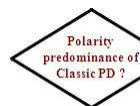
PD Interpretation



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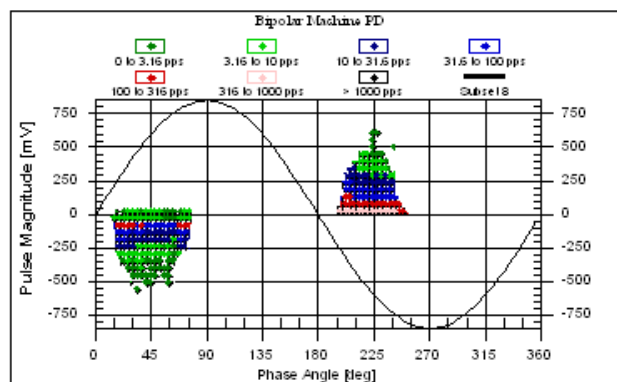
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Classic PD Pulses

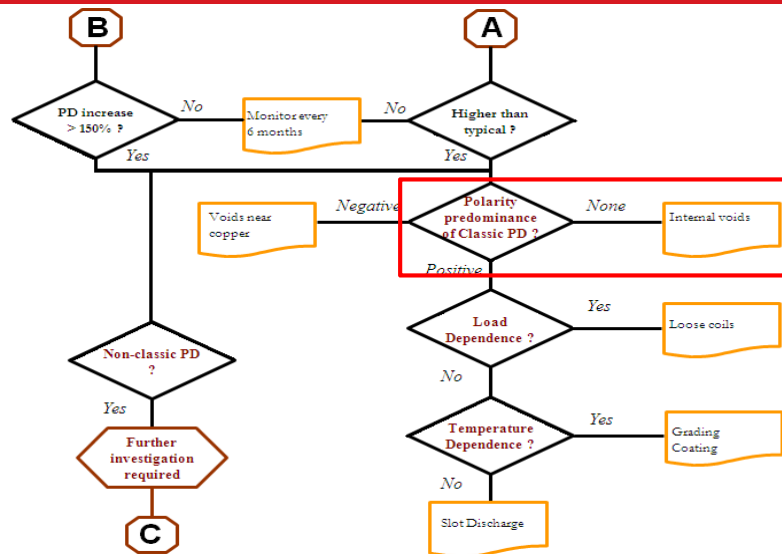
- Phase-to-ground
- Clumped at 45° and 225°



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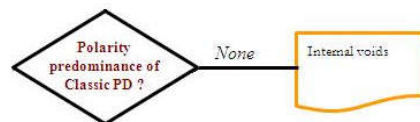
PD Interpretation



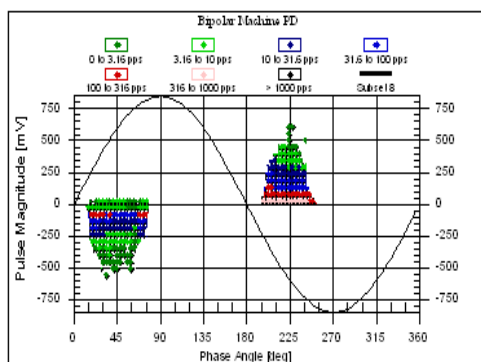
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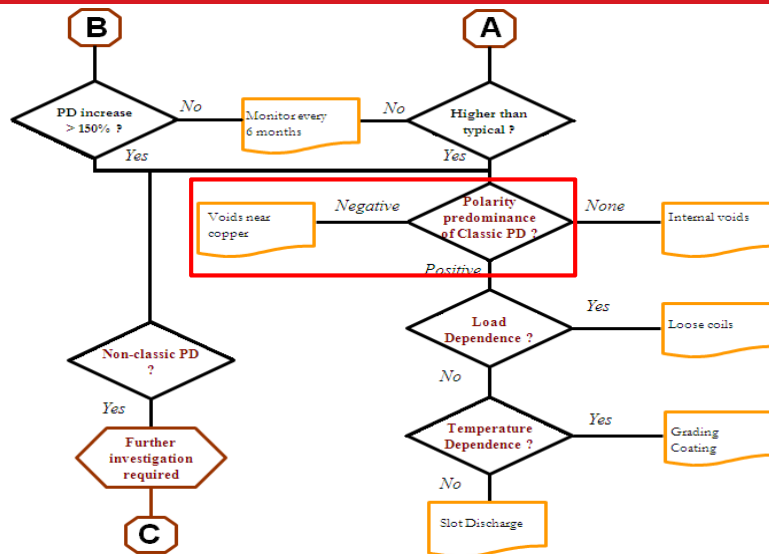
+/-	Load	Temp	Phase
No	No	Inverse	45°/225°



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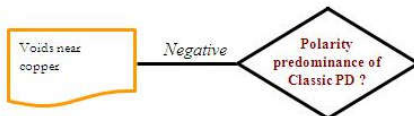
PD Interpretation



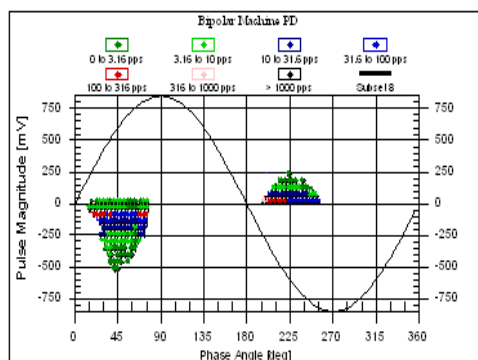
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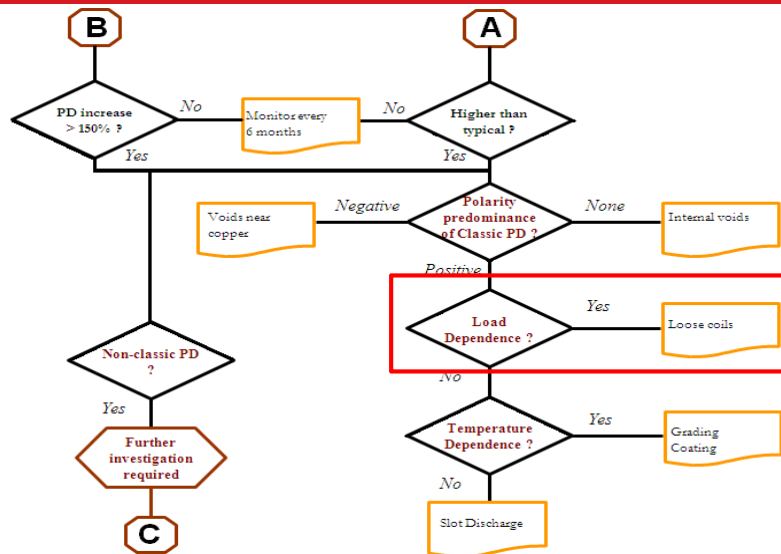
+/-	Load	Temp	Phase
Neg	No	Inverse	45°



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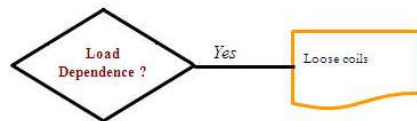
PD Interpretation



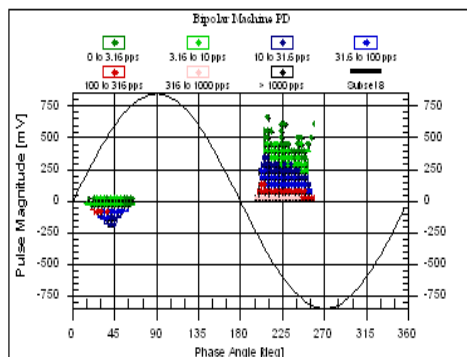
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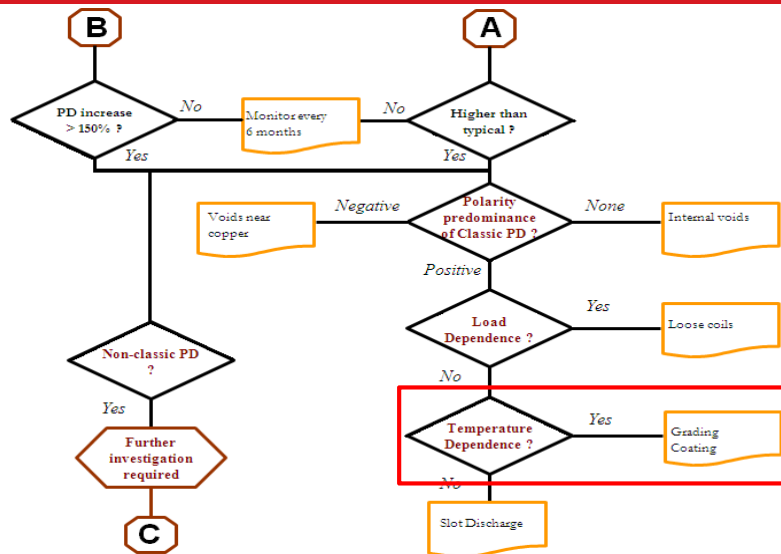
+/-	Load	Temp	Phase
Pos	Direct	Inverse	225°



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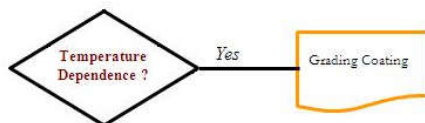
PD Interpretation



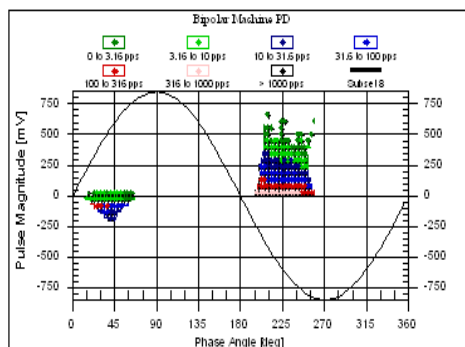
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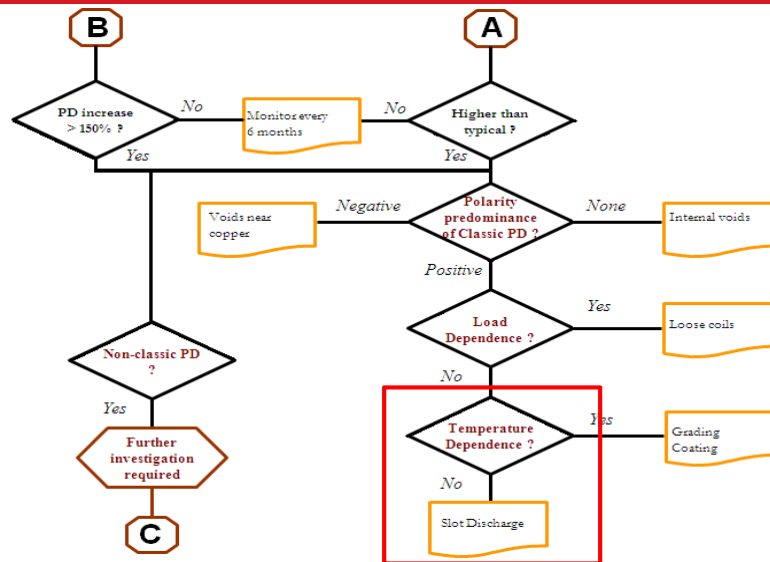
+/-	Load	Temp	Phase
Pos	No	Direct	225°



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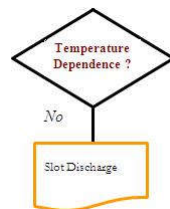
PD Interpretation



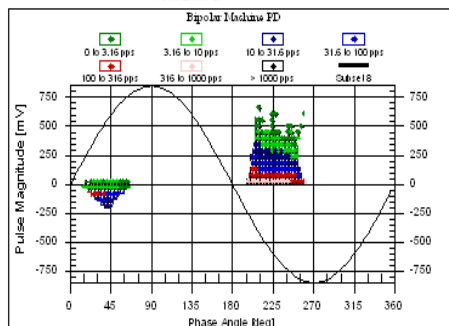
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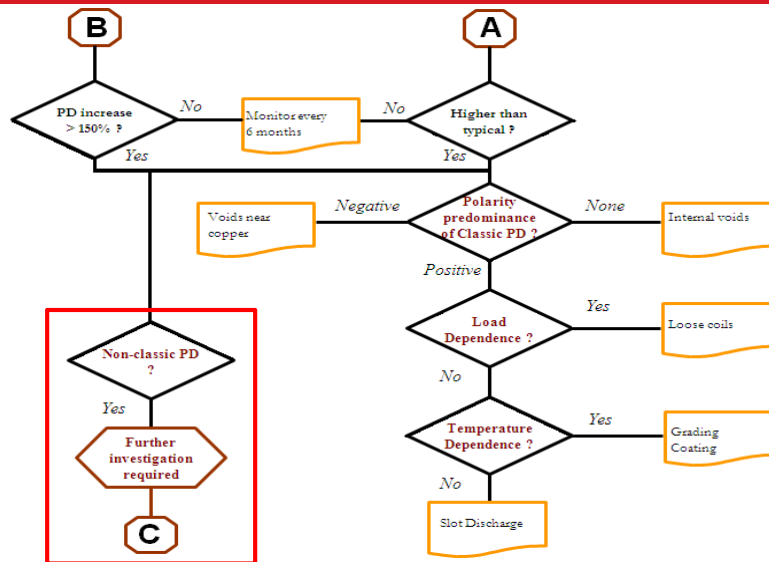
+/-	Load	Temp	Phase
Pos	No	No	225°



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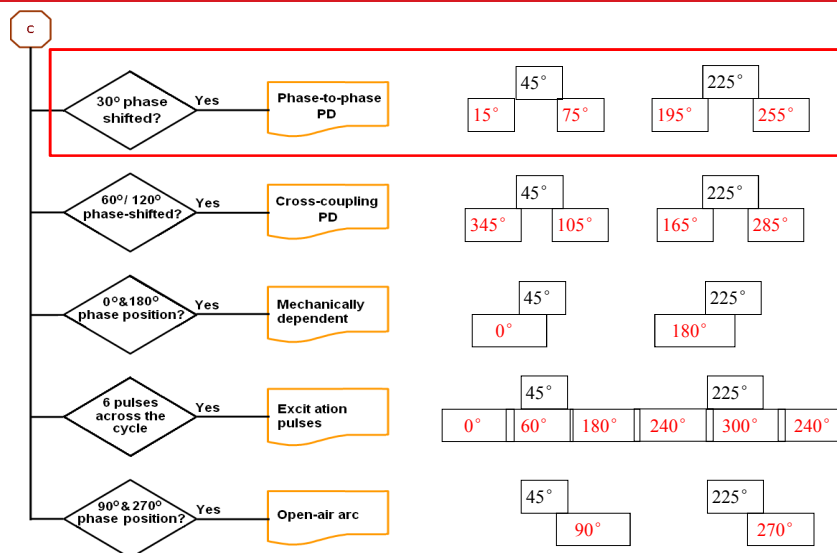
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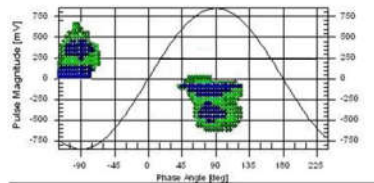
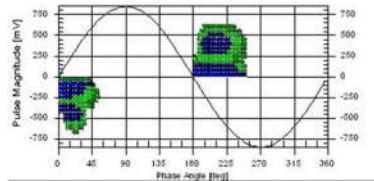
Yes

Phase-to-phase PD



Characteristics

- 30° phase-shifted
- Time domain
- Opposite polarity
- Similar in shape



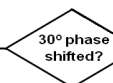
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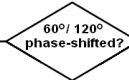


C



Yes

Phase-to-phase PD



Yes

Cross-coupling PD



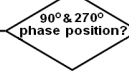
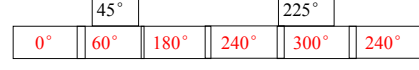
Yes

Mechanically dependent



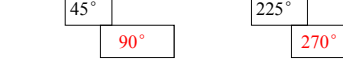
Yes

Excitation pulses



Yes

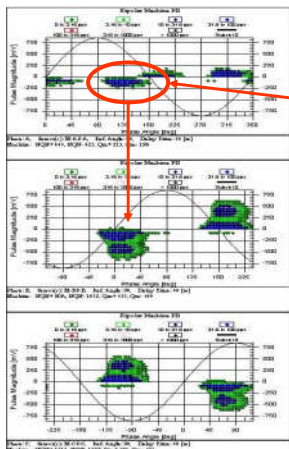
Open-air arc



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Characteristics

- Shifted on non-source phases
- Time domain
- Clusters may have opposite polarity
- Clusters are similar in shape
- Similar in shape

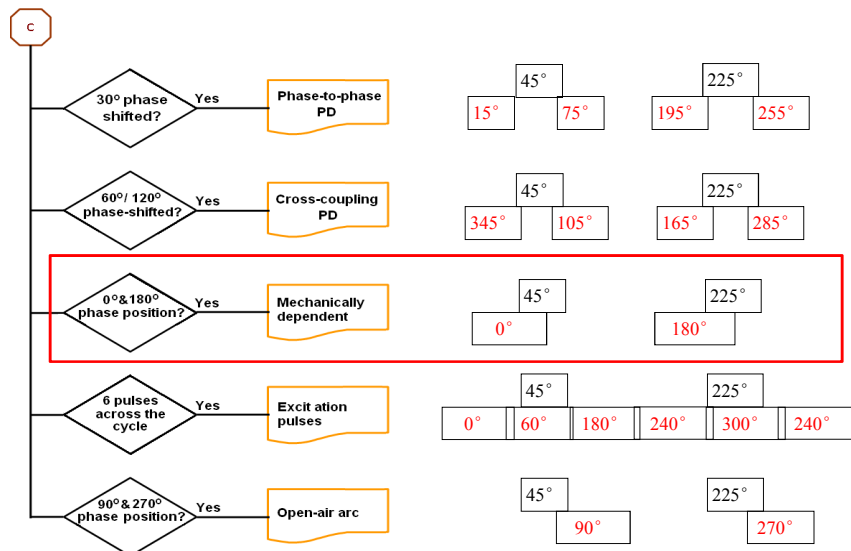
Cross-coupled from B phase



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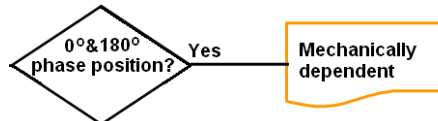
PD Interpretation



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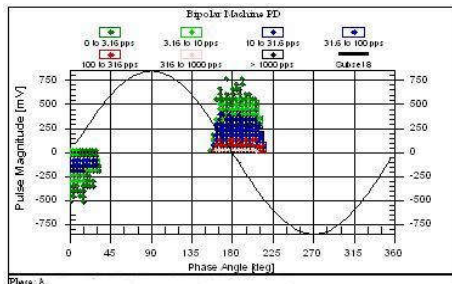
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PD from Mechanical Vibration

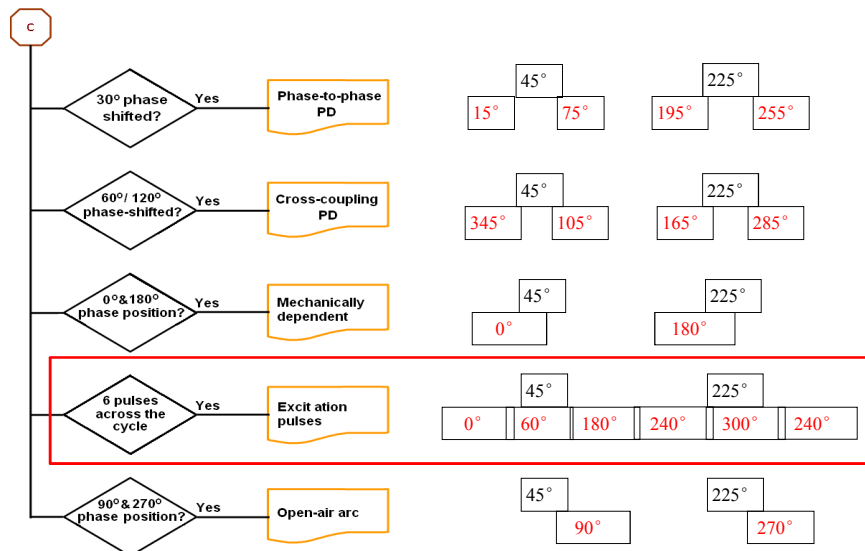
- PD clusters near 0° and 180°
- Problems such loose connections
- Arcing at the slot exit
- Vibration Sparking



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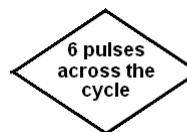
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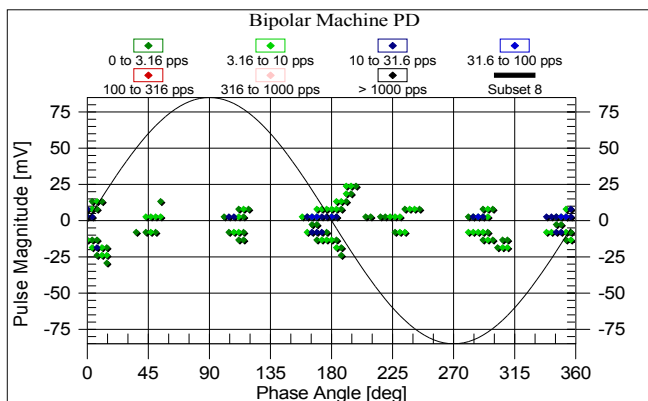


Yes

Excitation pulses

Characteristics

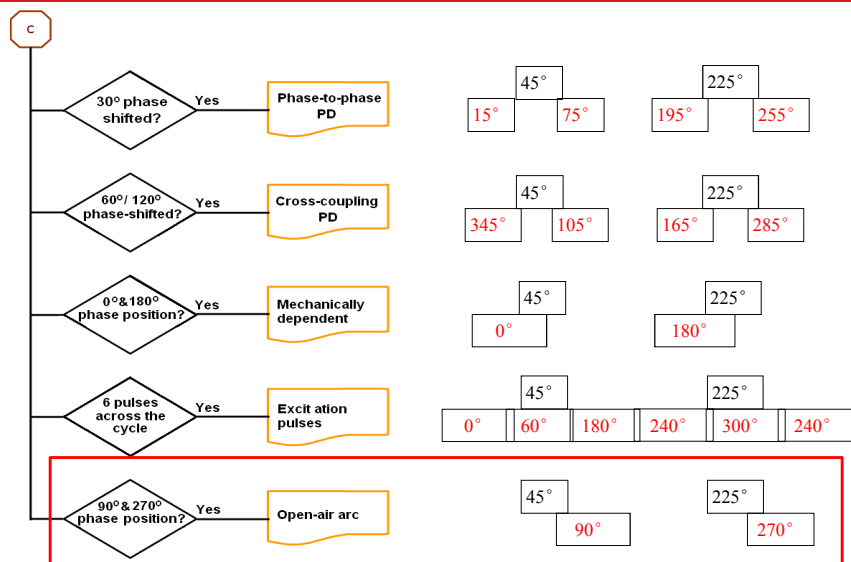
- Thyristor-firings static excitation
- Equal pulse repetitions
- Low magnitudes



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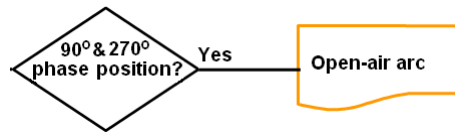
PD Interpretation



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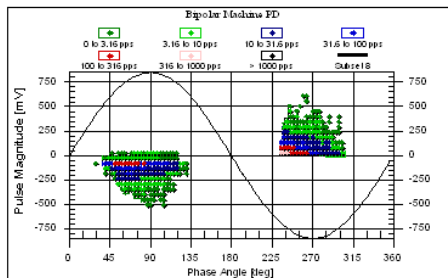
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PD Interpretation



PD on the surface

- PD pulses cluster near 90° and 270°
- Arcing along circuit ring bus
- Deterioration of voltage stress coatings



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Thanks for listening

Questions?

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